JC2066: IT Professionals: Ethical, Legal and Social Issues

L04: Privacy & Surveillance

Mar. 12th, 2026

MA, Jiawei Phoenix

Department of Computer Science

Where are the rest of the resources for this class?

On Canvas: https://canvas.cityu.edu.hk/courses/68163

Today's Agenda

1. **Introductory Provocations**
2. What is Privacy?
3. Privacy and Big Data
4. Approaches to data privacy
   - Anonymization
   - Encryption
   - Randomized Response
   - Differential Privacy
5. Large Language Model (LLM)

Who is this person?

Jeremy Bentham 1748-1832

JEREMY BENTHAM

PANOPTI WHAT?

PanoptiCam is a webcam mounted above Jeremy Bentham's **auto-icon**.

Learn **more** about this project!

UCL CENTRE FOR DIGITAL HUMANITIES

casa

BENTHAM PROJECT

UCL MUSEUMS

PANOPTICAM

UPDATES

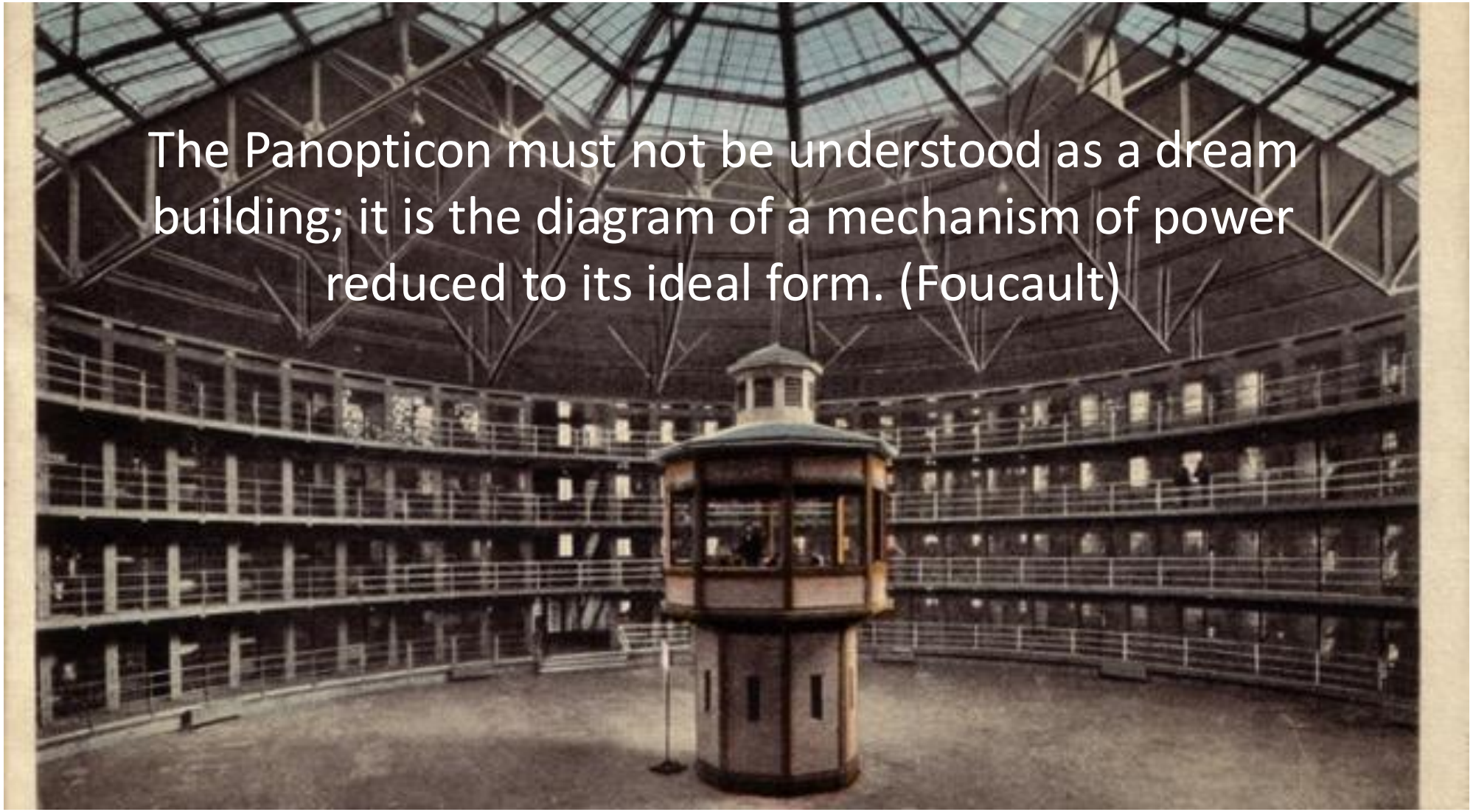
- Vote for us in the DH Awards!
- Some More Favourite PanoptiCam Views
- Jeremy! a message to you, Jeremy!
- The PanoptiCam in the Age of Digital Surveillance
- Some choice views from Bentham
- 'A great gogle eye': Some panopticon iconography
- Highlights of Jeremy's gaze so far...
- PanoptiCam is go!

Archive of Daily Timelapses

At the end of the South Cloisters of the main building of UCL stands a wooden cabinet, which has been a source of curiosity and perplexity to visitors. The cabinet contains the preserved skeleton of the philosopher and jurist Jeremy Bentham, (1748-1832), which dressed in his own clothes, and surmounted by a wax head, looks

Jeremy Bentham's 'perfect' prison | The Panopticon

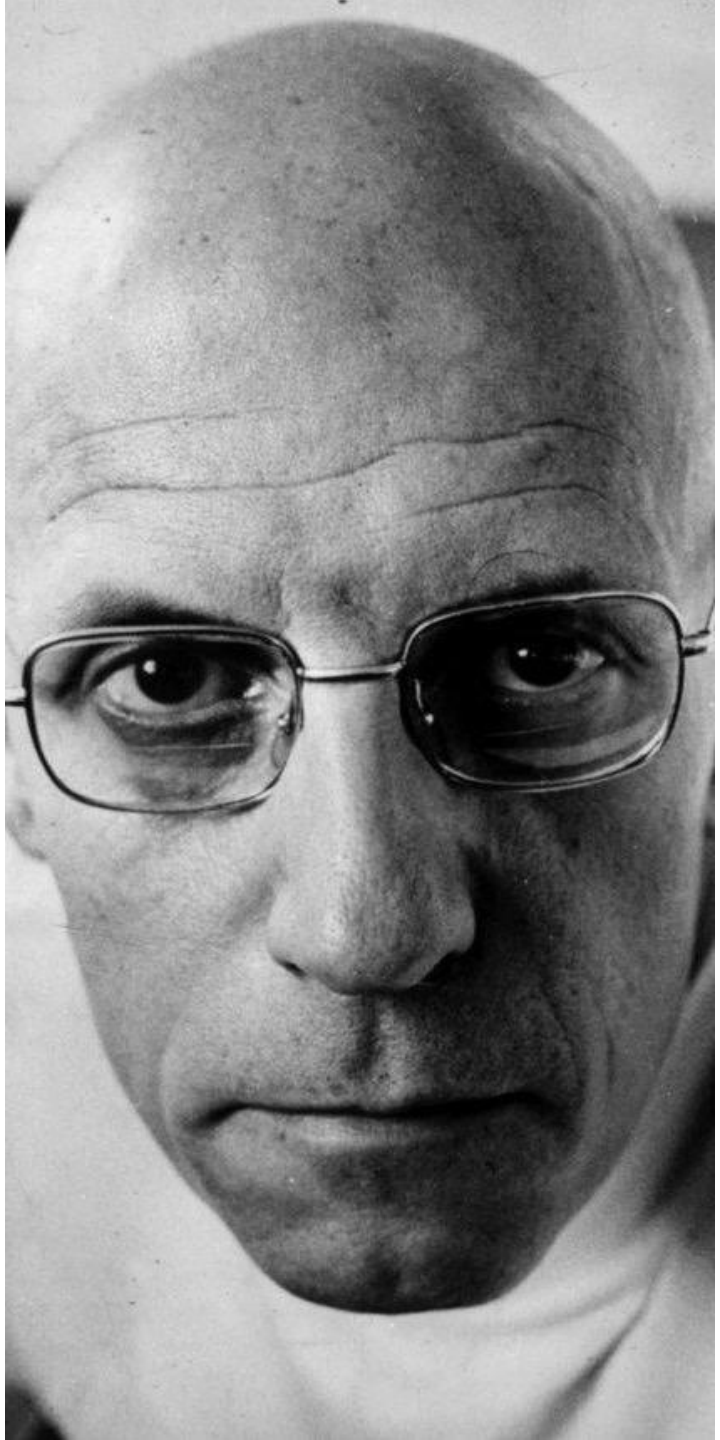
Stateville Correctional Center, Illinois



Michel Foucault (French philosopher)

- He is seen, but he does not see; he is an object of information, never a subject in communication.

Discipline and Punish



A Modern Panopticon?

The Digital Panopticon

Citizens can sign up as Virtual Texas Deputies to participate in border surveillance through this social network. Virtual Texas Deputies from around the country will monitor the streaming video from these cameras 24/7 and report any suspicious activities directly to the Border Sheriffs via email. All emails regarding suspicious activity will be submitted anonymously.

SIGN IN SHOP DONATE

TS & LIFE MUSIC SHOWS & PODCASTS SEARCH

NATIONAL

A New Way To Patrol The Texas Border: Virtually

February 23, 2009 · 12:56 PM ET
Heard on All Things Considered

JOHN BURNETT

In a controversial program aimed at enhancing border security, Texas sheriffs have erected a series of surveillance cameras along the Rio Grande and connected them to the Internet.

A video on blueservo.net shows Texan law enforcement driving a raft on the Rio Grande away from the U.S. border. According to the site, a "Virtual Texas Deputy" spotted the raft.
blueservo.net

Thousands of people are now virtual Border Patrol agents — and they're on the lookout for drug smugglers and illegal immigrants.

Robert Fahrenkamp, a truck driver in South Texas, is one of them.

After a long haul behind the wheel of a Peterbilt tractor-trailer, he comes home, sets his 6-foot-6-inch, 250-pound frame in front of his computer, pops a Red Bull, turns on some Black Sabbath or Steppenwolf, logs in to

Facial Recognition

The New York Times

Account

The Secretive Company That Might End Privacy as We Know It

A little-known start-up helps law enforcement match photos of unknown people to their online images — and "might lead to a dystopian future or something," a backer says.

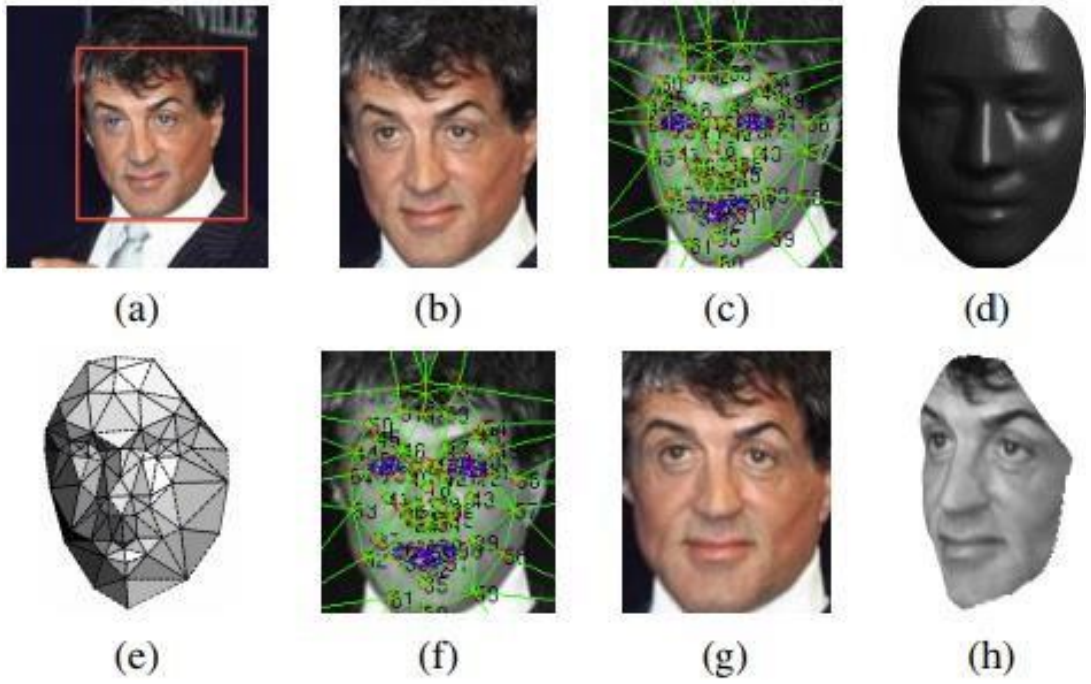
Facial Recognition

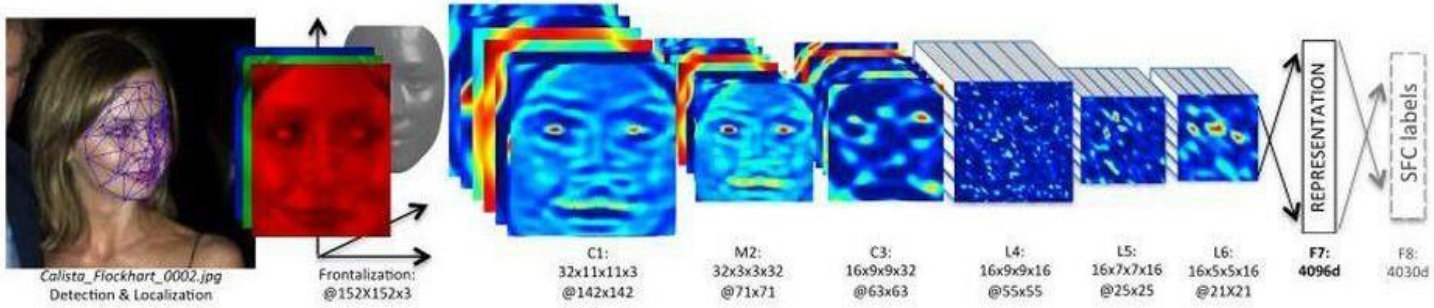
- Applications
  - Verification (e.g., unlocking you iPhone with FaceID)
  - Tagging (e.g., Facebook photos)
    - Facebook recently announced it will no longer tag photos
  - Many others – we'll talk about this in case study in sections this week

- Surveillance
  - Chinese police use facial recognition at concert with 60,000 people to identify and arrest suspect
  - Facial recognition used at Taylor Swift concerts to identify stalkers

Facebook's DeepFace

- Facebook DeepFace algorithm (Taigman et al, 2014)
  - Nine layer neural network with 120+ million weights
  - Trained on 4+ million images
  - Build 3D model from 2D image to get "frontal" view
  - Claim 97.35% accuracy
  - Comparable to humans





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What is Privacy?

The most striking thing about the right to privacy is that nobody seems to have any very clear idea what it is.

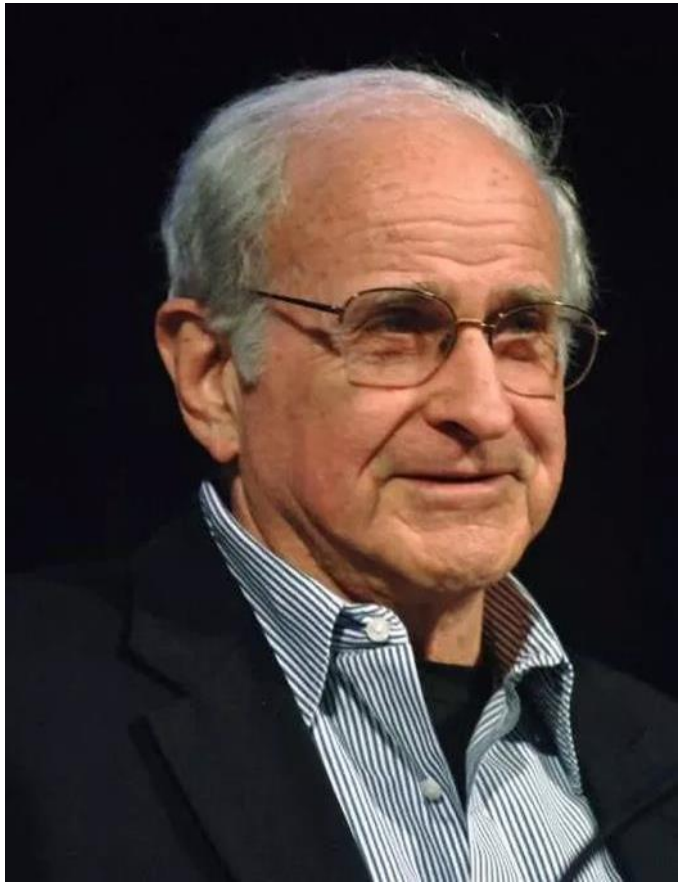
Judith Jarvis Thomson, 1975 MIT Philosopher, “The Right to Privacy”

Privacy is a concept in disarray.

Daniel Solove Georgetown Law Professor

Few values so fundamental to society as privacy have been left s undefined in social theory or have been the subject of such vague and confused writing by social scientists.

Alan Westin, 1967



What is Privacy?

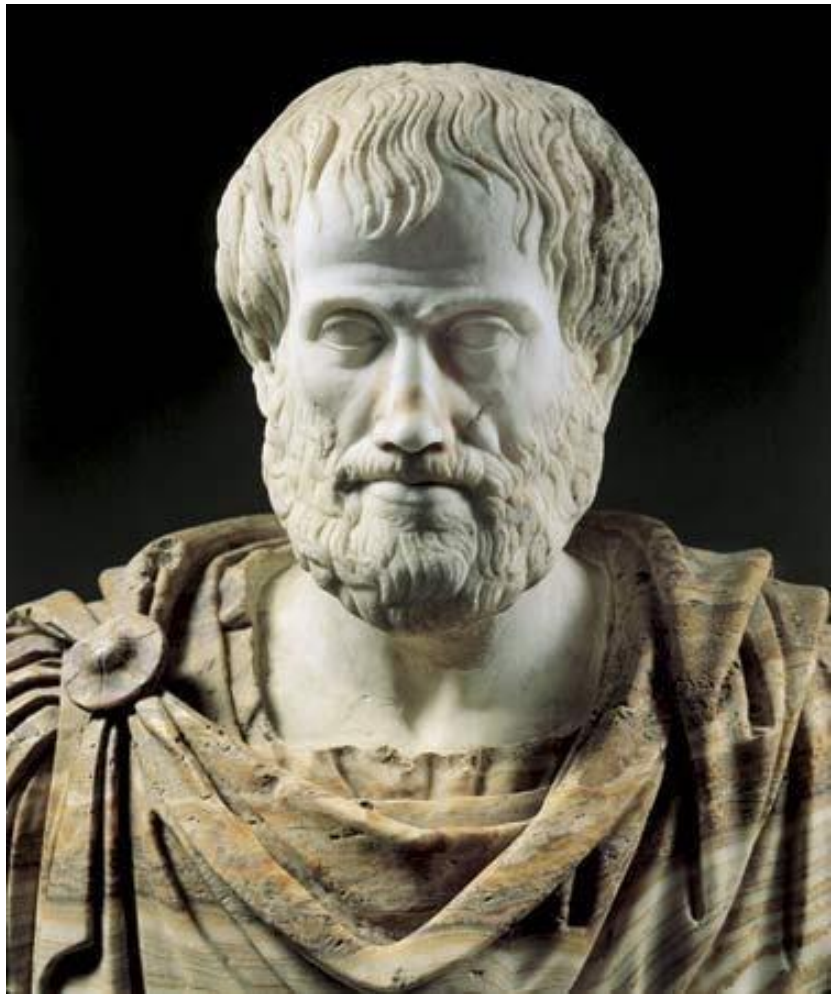
Aristotle provides us an early view:

The Public Sphere: politics and commerce

X

The Private Sphere: domestic life, family and friends

Individuals have interests in shielding the public (politics and commerce) from seeing/hearing what happens in private.



This Photo by Unknown Author is licensed under CC BY-NC

What is Privacy?

But the public/private sphere distinction is crude.

Social context is what's important, whether in the public or private sphere (Nissenbaum). We want to be able to control *what is known about us*, *how we present ourselves*, whether we are in public or private spheres.

Individuals have privacy interests even when in public!

What is Privacy?

Privacy is multi-dimensional (Solove)

Privacy concerns control over information about ourselves (self-determination), and about the pursuit of relationships with others (intimacy).

- *Individuals have privacy interests even when in public!*

Privacy is not about hiding things that are wrong or embarrassing.

It is about protecting intimacy, freedom, and control.

(Solove, “Nothing to Hide Argument”)

What is Privacy?

Modern view is concerned with both private and public spheres

Privacy is central to *individual autonomy* or *self-determination*:

> ***The claim of individuals to determine for themselves when, how, and to what extent information about them is shared with or communicated to others.***

Violations of privacy impose ***harms*:**

Loss of freedom, intimacy, control over self and future.

Four Categories of Privacy Harms

1. **Information Collection** (e.g., surveillance)
2. **Information Processing** (e.g., aggregation and inferences made from big data)
3. **Information Dissemination**
   A. Breach of confidentiality
   B. Disclosure to third parties
   C. Blackmail
4. **Invasion of Privacy**
   A. Intrusion
   B. Decision interference

(Solove, A Taxonomy of Privacy)

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Privacy and Big Data

- When interacting with digital devices and the web, individuals give up information about themselves, creating *digital exhaust streams*.

- Then algorithms can make **inferences and predictions** about people from these data.

Predictive Analytics at Target (and every other big company)

- **Data Collected by Target on shoppers, all linked to unique "Guest IDs":**

- Purchases made
- Coupons used
- Credit cards used/owned
- Surveys completed
- Emails opened
- Website visits
- Other websites visited
- Customer service calls

Plus data purchased from data brokers or acquired through surveys:

- Age
- Marriage status
- Income
- Zip code
- Race/ethnicity/religion
- Alma mater
- Magazine/newspaper subs
- Job history
- And much more

Create a “Pregnancy Prediction” Score

As Pole’s computers crawled through the data, he was able to identify about 25 products that, when analyzed together, allowed him to assign each shopper a “pregnancy prediction” score. More important, he could also estimate her due date to within a small window, so Target could send coupons timed to very specific stages of her pregnancy.

One Target employee I spoke to provided a hypothetical example. Take a fictional Target shopper named Jenny Ward, who is 23, lives in Atlanta and in March bought cocoa-butter lotion, a purse large enough to double as a diaper bag, zinc and magnesium supplements and a bright blue rug. There’s, say, an 87 percent chance that she’s pregnant and that her delivery date is sometime in late August. What’s more, because of the data attached to her

TARGET OFFER
VALID 1/26/20–2/1/20
$20
GiftCard
Free $20 Target GiftCard™
with select baby department purchase of $100 or more
• Wipes • Diapers & training pants • Infant formula
• Baby food • Baby toiletries
Find this offer in Wallet in the Target app

Luvs
Huggies
Pampers
Pull-Ups
FreeStuffFinder.com

Aggregation and Inference

- Aggregation of public information
  - Each time you are outside, that's essentially public information
  - Aggregation: someone following you outdoors all the time

  - Purchasing something in store, people next to you can see what you buy
  - Aggregation: compilation of everything you have bought at all stores

  - So at what point do we feel uncomfortable?

- In the on-line world, aggregation and inference is the standard
  - How well you can do it is the difference between Yahoo and Google

But I Just Pressed the Like Button

- Kosinski et al (2013) analyze "Likes" on Facebook to see what they can infer about you
  - 58,466 volunteers provided "Likes", demographics, and psychometrics
  - Average of 170 "Likes" per person
  - Built predictive models for various attributes based only on "Likes"
  - Even one "Like" carries predictive information!

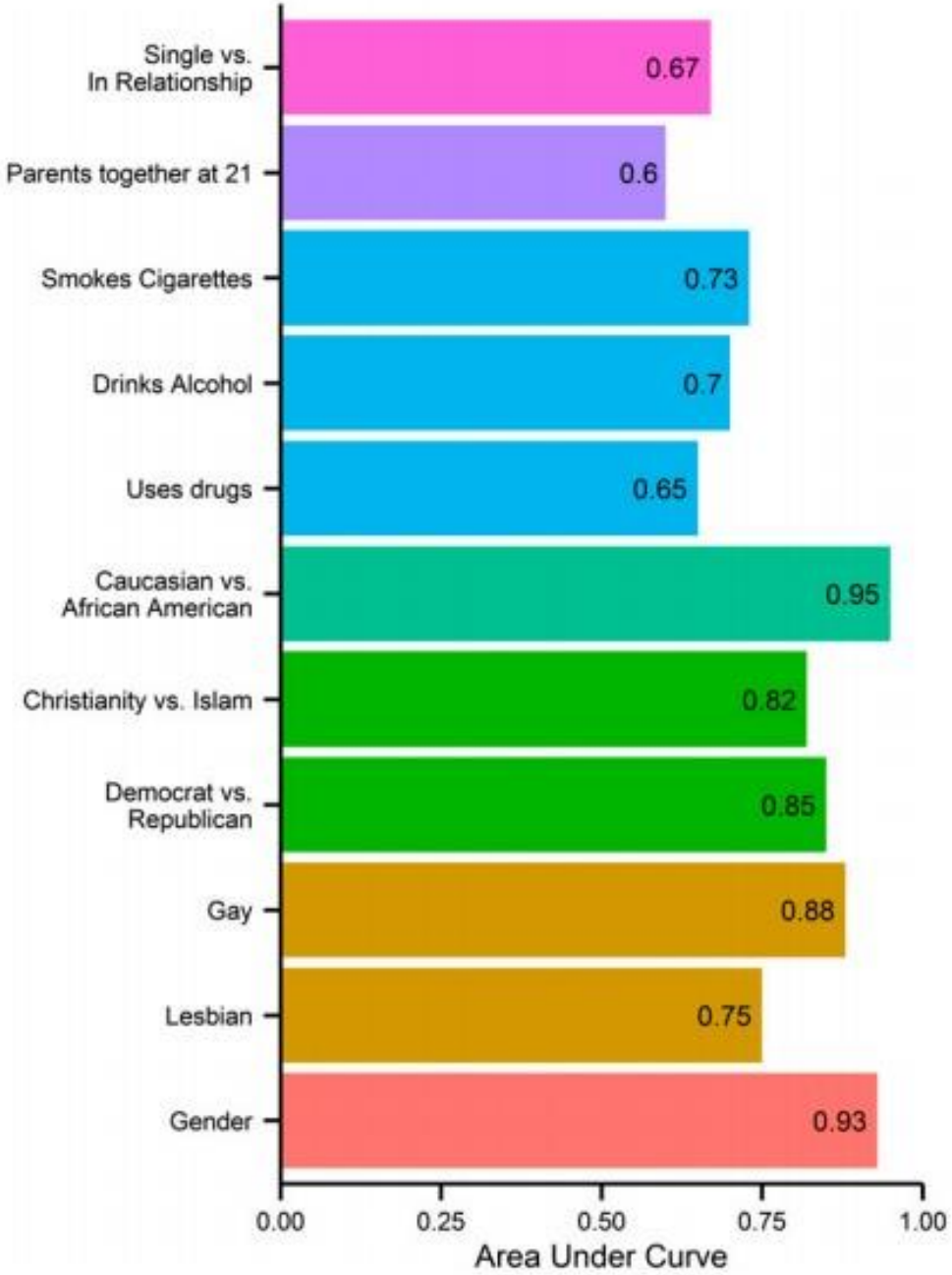

Fig. 2. Prediction accuracy of classification for dichotomous/dichotomized attributes expressed by the AUC.

An Information Ecosystem…

- LiveRamp/Acxiom
  - $3B company headquartered in San Francisco
  - Provide data for marketing segmentation, which are even named
  - Let's look at their "Life Stage Clustering System: 'PersonicX'"

- "Full Steaming" is a mix of affluent, well-educated couples and singles that have a net worth exceeding $500,000…

- "Lavish Lifestyles" contains established couples with teenage kids, minivans and mortgages…

- "Tots & Toys" is dominated by affluent and well-educated working couples with preschool-age children. They are homeowners, mainly in single- family houses…

…Where Everyone is Classified…

- At a mean age of 25, “Early Parents” represents one of the youngest of the segments. It contains single and married parents in their mid-20s whose spending habits and leisure time are heavily influenced by their young children. Early Parents ranks among the nation’s lowest clusters for income and net worth…
- “Truckin’ & Stylin’” households are in their mid- to late-30s and live in rural towns. They rank just below average for household income but drop to the bottom of the list (66th) for net worth…
- “Resilient Renters” represents an ethnically diverse group of singles. They are renters and, if employed, earn extremely low wages in clerical and blue-collar jobs. This cluster represents one of the lowest for net worth…
- “Metro Parents” is a group struggling with single parenthood and the stresses of urban life on a small budget…

…Including You, “Collegiate Crowd”

- With a mean age of 21, this group represents the youngest of all the clusters. The cluster has a high concentration of students, a correlating low income and net worth, and high mobility.

- Collegiate Crowd is made up of single, highly mobile apartment dwellers. Over half of all Collegiate Crowd live in the big-three Central regions, 82% in metropolitan areas boasting colleges and universities. The group has a 31% adult concentration of students, and ranks 53rd for household income and 50th for net worth.

- Not surprisingly, they are twice as likely to have education loans. This group is constantly online using laptops and the Internet for news, weather and job hunting. As for activities, volleyball, basketball, barhopping and movie going fill their time, and this group is much more likely than average to attend college football games.

The Information Ecosystem

- Are there any benefits here?

- Consider the example of Target building a predictive model to identify pregnant women
  - What if this was a government trying to reach women to provide:
    - Information about and access to prenatal health care
    - Nutritional information and supplemental nutrition assistance

  - Also allows for longer-term tracking
    - When child is toddler (pre-K education)
    - When child goes to school (school options/vouchers)
    - When child might prepare to go to college (financial aid)
    - When child becomes an adult

A Digression: Ad Tech and You

- This is the surveillance capitalism economy.
- This is ad tech.
- This is what many jobs at Facebook, Google, Amazon are.

"The best minds of my generation are thinking about how to make people click ads."

--Jeff Hammerbacher, a 28-year-old Silicon Valley tech whiz who went from being an early employee at Facebook to co-founding the data analysis start-up Cloudera, in Ashlee Vance's *BusinessWeek* story about the advertising and social-media driven bubble in Silicon Valley.

Protecting or Violating Privacy?

Question:

- *Should privacy be treated differently depending on whether it is the **government** or a **company** that collects data?*

- *Different privacy standards for different agents?*

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Perspectives on Data Privacy

- Data privacy often involves a balance of competing interests
  - Privacy vs. national security/personal safety/innovation/convenience
- Making data available for meaningful analysis
  - For the public good
    - Auditing algorithmic decision-making for fairness
    - Medical research and health care improvement
    - Protecting national security
  - For the private good
    - Personalized advertising

Perspectives on Data Privacy

- Protecting individual privacy
  - Personal value of privacy and respect for individual
  - Freedom of speech and activity
  - Avoiding discrimination
  - Regulation/laws: FERPA, HIPAA, GDPR, etc.
  - Preventing access from “adversaries”

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Anonymization

- Basic idea: drop personally identifying features from the data

Employer	Job Title	Nationality
Stanford University	Professor	Iran
Google Inc.	Senior Engineer	USA

D.O.B	Zipcode	Has Condition?
May 10,	94306	No
June 6,	94040	Yes

- What if we also drop gender and birth year?
  - Just to be safe, since gender and age are protected characteristics
- How well does that work?

Massachusetts Hospital Visits

- Massachusetts Group Insurance Commission (GIC) released "anonymized" data of state employee hospital visits
  - 135,000 data records

- “William Weld, then Governor of Massachusetts, assured the public that GIC had protected patient privacy by deleting identifiers.” (Paul Ohm, 2015)

Massachusetts Hospital Visits

Latanya Sweeney, now Professor at Harvard, then graduate student at MIT, investigates…

- Sweeney knew Weld lived in Cambridge, MA
- For $20, she bought Cambridge voter list, containing: name, address (including zip code), birth date, and gender of 54,805 voters in city
- She joined voter data with GIC data, de-anonymizing (re-identifying) Weld
- Only six people in Cambridge shared Weld's birth date, only three of those six were men, and only Weld lived in his zip code
- She sent Weld's health records to his office

De-anonymization Can Be Easy

- Sweeney (2000): 87 percent of all Americans could be uniquely identified using: zip code, birthdate, and sex.
  - ~42,000 zip codes
  - 365 birthdates per year for ~80 years
  - Most individuals identify as one of two sexes
  - 42,000 * 365 * 80 * 2 ≈ 2.5B. Population of US ≈ 330M
- Try it yourself: Data Privacy Lab: https://aboutmyinfo.org/identity

Fill out the form below to see how unique you are, and therefore how easy it is to identify you from these values.
Please note that this service is still under development.

Date of Birth May 10 1970

Gender Male Female

ZIP Code 94306

ZIP code must be 5 digits long.

Submit →

Your Profile

Gender: Male
ZIP Code: 94306 (pop. 26469)

Date of Birth	5 / 10 / 1970	Easily identifiable by birthdate (about 1).
Birth Year	1970	Lots with your birth year (about 189).
Range	1970 to 1974	Lots in the same age range as you (about 947).

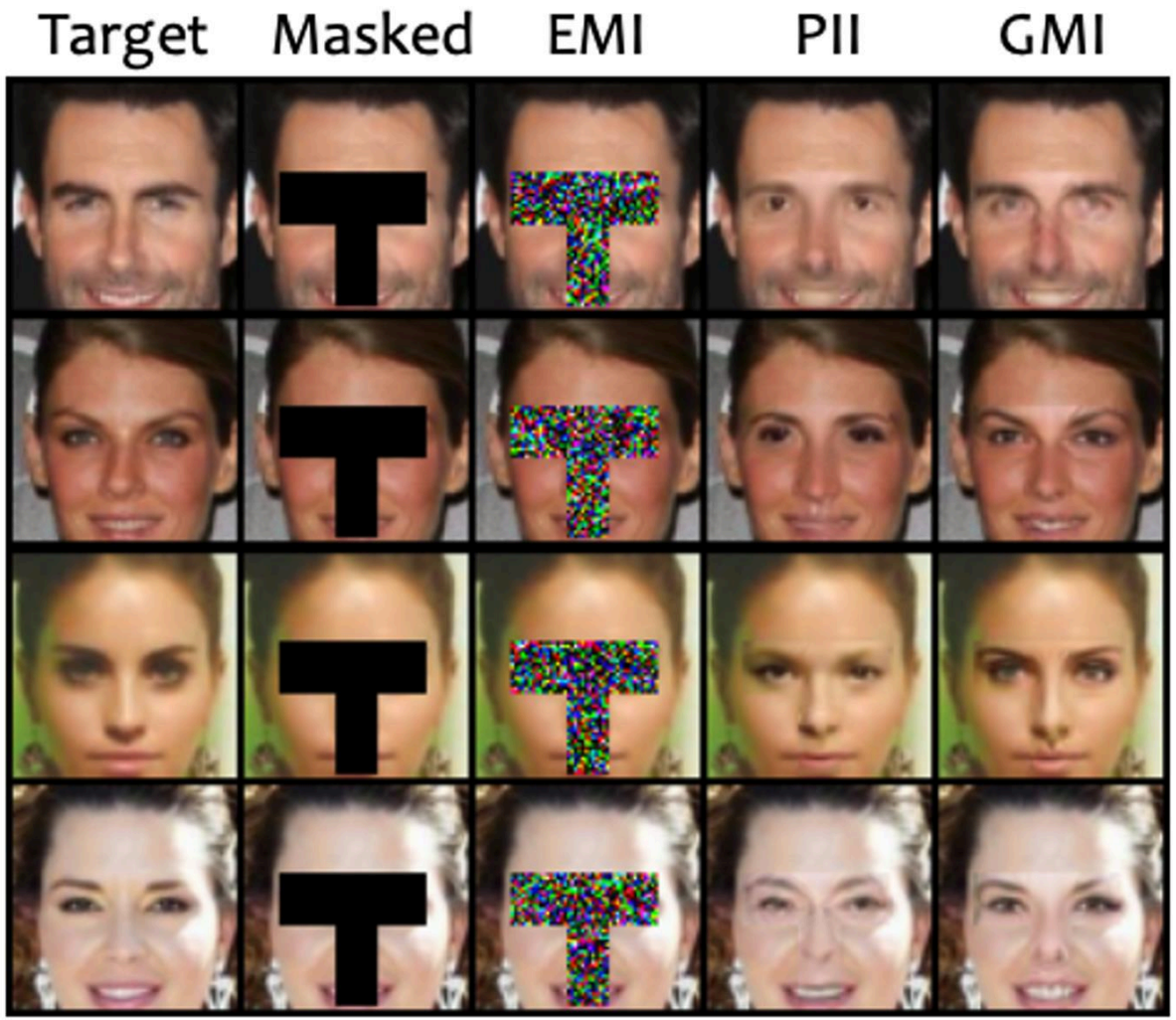
De-anonymization Can Be Easy (Image Model)

Even if you dont release the raw data, the weights of a trained network might reveal sensitive information.

Model inversion attacks recover information about the training data from the trained model.

Heres an example of reconstructing individuals from a face recognition dataset, given a classier trained on this dataset and a generative model trained on an unrelated dataset of publicly available images.

Col 1: training image. Col 2: prompt. Col 4: best guess from only public data. Col 5: reconstruction using classi cation network.



Zhang et al., The secret revealer: Generative model-inversion attacks against deep neural networks. https://arxiv.org/abs/1911.07135

De-anonymization Can Be Easy (Language Model)

A neural net language model trained on Linux source code learned to output the exact text of the GPL license.

```
/*
 *  Copyright (c) 2006-2010, Intel Mobile Communications.  All rights reserved.
 *
 *   This program is free software; you can redistribute it and/or modify it
 * under the terms of the GNU General Public License version 2 as published by
 * the Free Software Foundation.
 *
 *        This program is distributed in the hope that it will be useful,
 * but WITHOUT ANY WARRANTY; without even the implied warranty of
 *   MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
```

Gmail uses language models for email autocompletion. Imagine if the autocomplete feature spits out the entire text of one of your past emails.

And People Get Fired For It

- America Online (AOL) releases an anonymized log of search queries for research purposes in 2006
  - 20M searchers
  - 650,000 users (tagged by “anonymous” ID)
- NY Times de-anonymizes several individuals in data
  - User #4417749: Thelma Arnold, a 62-year-old widow from Lilburn, GA
  - Queries included:
    “landscapers in Lilburn, Ga” (several people with last name *Arnold*) “homes sold in shadow lake subdivision gwinnett county georgia” “60 single men” “dog that urinates on everything”
- AOL Chief Technology Officer resigns, other researchers fired

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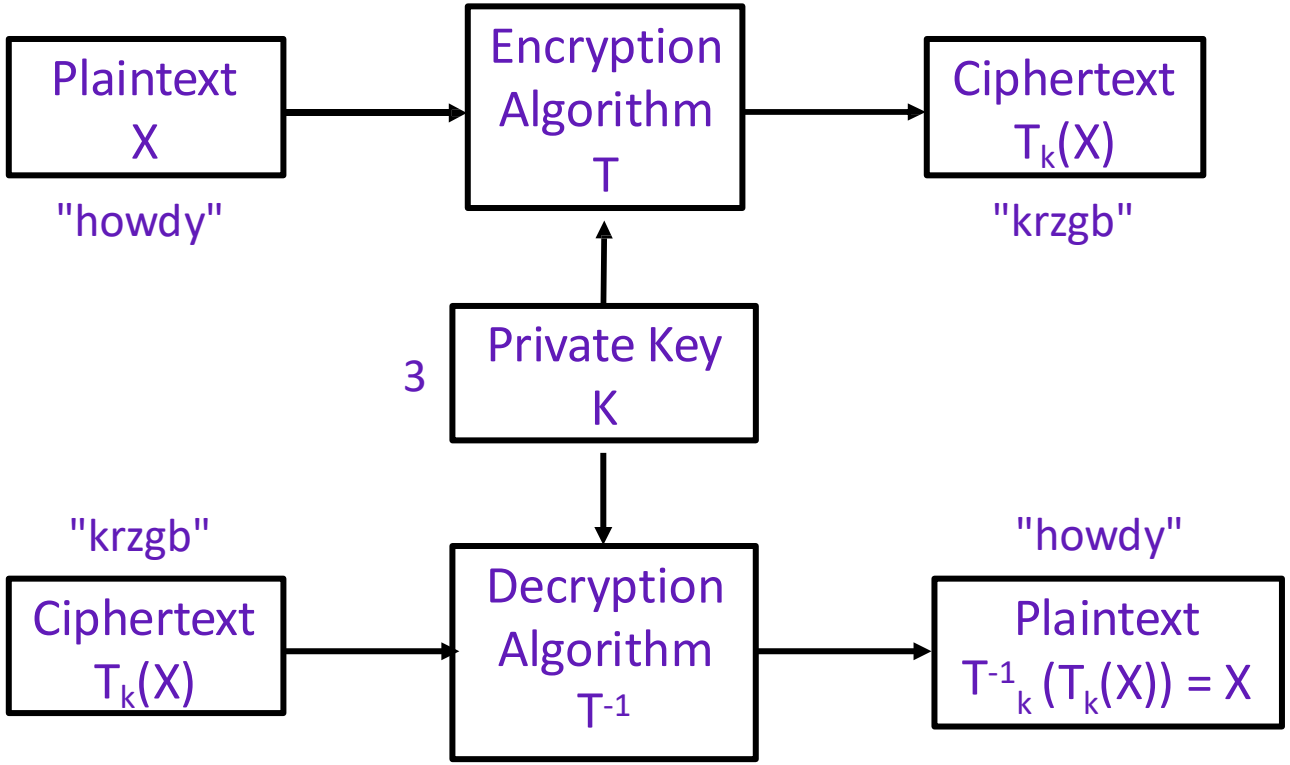
Encryption

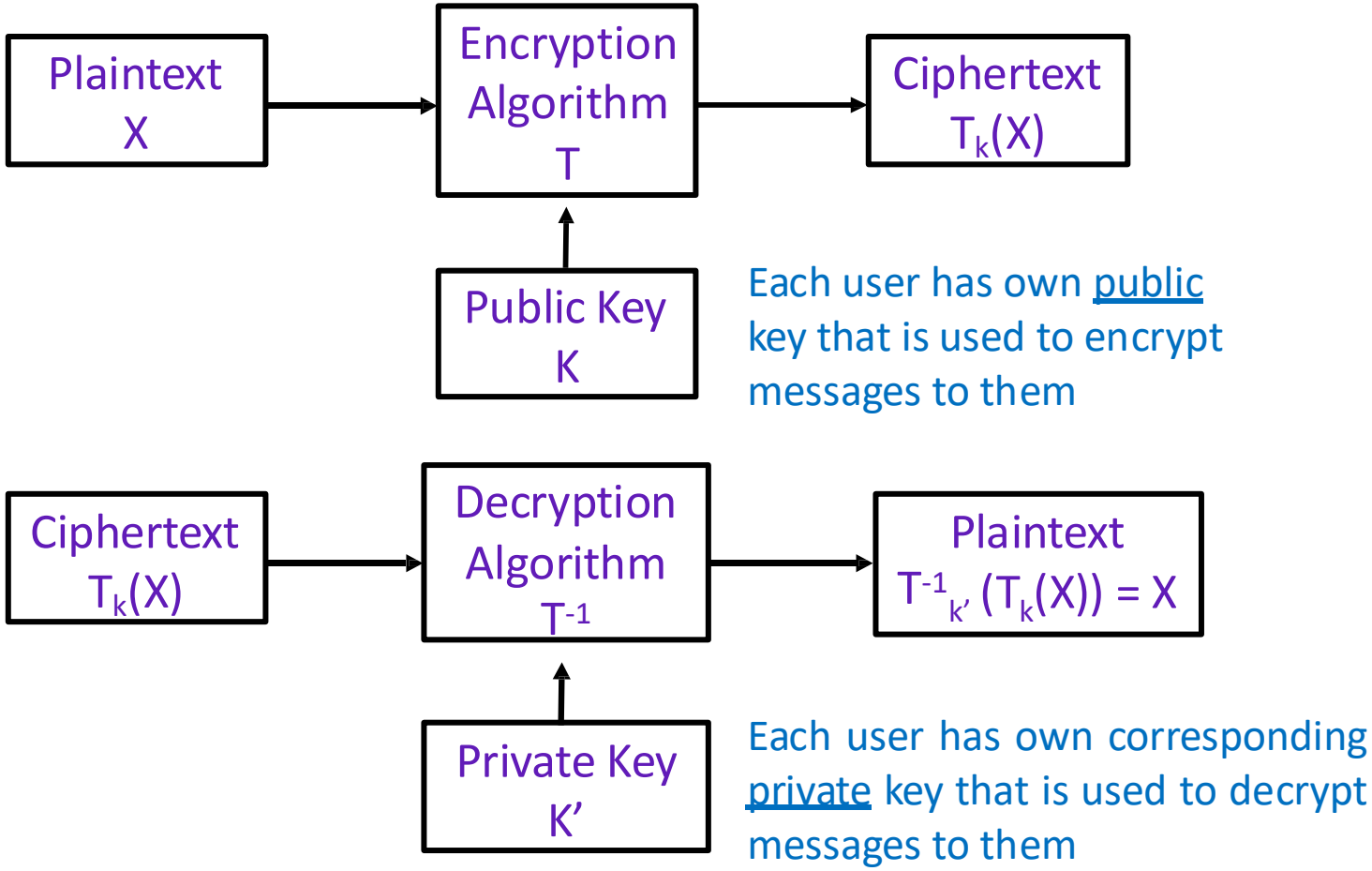
- Encryption: process of **encoding** information so that only those with authorization can access it
- Why encrypt?
  - Keep data private/secure (e.g., iPhone)
  - Prevent eavesdropping (e.g., WhatsApp end-to-end encryption)
  - Guarantee data not altered in transport (e.g., database integrity)
- Data recipient must be able to decrypt data for analytics
  - To share healthcare data for analytics, recipient must decrypt data
- Research direction: fully homomorphic cryptosystems
  - Allow arbitrary mathematical operations to be performed on *encrypted* data

Symmetric Key Cryptography



- Example: Caesar Cipher
  - Encrypt: shift letters *forward* K places in alphabet, wrapping from ‘Z’ to ‘A’
  - Decrypt: shift letters *backward* K places, wrapping from ‘A’ back to ‘Z’

Public Key Cryptography



- Public key and private key somehow related to each other to make this possible

Public Key Cryptography

- 1976: Public key cryptography proposed by Diffie and Hellman
  - Initial proposal proved to be insecure
- 1978: Rivest, Shamir, and Adleman propose RSA algorithm
  - Secure (as far as we know) version of public key cryptography
- RSA cryptosystem
  - Public key (n) is based on product of two prime numbers p and q, $n = pq$
  - Private key (p and q) is based on knowing factorization of n into p and q
    - Factoring large numbers is hard, and our security relies on that!
  - Messages in this system represented as numbers
- Just to keep you up at night: practical quantum computing could break many common currently used cryptosystems (like RSA)

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Randomized Response

Intuition: Randomized response is a survey technique that ensures some level of privacy.

Example: Have you ever dodged your taxes?

- Flip a coin.
- If the coin lands Heads, then answer truthfully.
- If it lands Tails, then flip it again.
  - If it lands Heads, then answer Yes.
  - If it lands Tails, then answer No.

Probability of responses:

	Yes	No
Dodge	0.75	0.25
No Dodge	0.25	0.75

Randomized Response

Tammy the Tax Investigator assigns a prior probability of 0.02 to Bob having dodged his taxes. Then she notices he answered Yes to the survey. What is her posterior probability?

$$\Pr(\text{Dodge} \mid \text{Yes}) = \frac{\Pr(\text{Dodge})\Pr(\text{Yes} \mid \text{Dodge})}{\Pr(\text{Dodge})\Pr(\text{Yes} \mid \text{Dodge}) + \Pr(\text{NoDodge})\Pr(\text{Yes} \mid \text{NoDodge})} = \Pr(\text{Yes})$$

$$= \frac{0.02 \cdot \frac{3}{4}}{0.02 \cdot \frac{3}{4} + 0.98 \cdot \frac{1}{4}} \approx 0.058$$

So Tammy's beliefs haven't shifted too much.

More generally, randomness turns out to be a really useful technique for preventing information leakage.

Randomized Response

How accurately can we estimate μ the population mean?

Let $X_T^{(i)}$ denote individual i's response if they respond truthfully, and $X_R^{(i)}$ individual i's response under the RR mechanism.

With maximum likelihood estimate, if everyone responds truthfully:

$$\widehat{\mu_T} = \frac{1}{N}\sum_{i=1}^{N} X_T^{(i)}$$

But how to estimate μ from the randomized responses $\{X_R^{(i)}\}$?

$$\mathbb{E}\left[X_R^{(i)}\right] = \frac{1}{4}(1-\mu) + \frac{3}{4}\mu$$

$$\widehat{\mu_R} = \frac{2}{N}\sum_{i=1}^{N} X_R^{(i)} - \frac{1}{2}$$

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Differential Privacy

Basic setup:

- There is a database D which potentially contains sensitive information about individuals.
- The database curator has access to the full database. We assume the curator is trusted.
- The data analyst wants to analyze the data. She asks a series of queries to the curator, and the curator provides a response to each query.
- The way in which the curator responds to queries is called the mechanism. We'd like a mechanism that gives helpful responses but avoids leaking sensitive information about individuals.

Intuition: Consider two databases, one without your data and one with your data (i.e., one extra row of data). Result of a statistical query should be almost indistinguishable between two databases.

Alternative intuition: a subject with data in the database is subject to no more harm from data analysis than if they were not in the database

Differential Privacy

Two databases D_1 and D_2 are neighboring if they agree except for a single entry.

Idea: if the mechanism behaves nearly identically for D_1 and D_2, then an attacker can't tell whether D_1 and D_2 was used (and hence can't learn much about the individual).

Definition: A mechanism M is ε-differentially private if for any two neighboring databases D_1 and D_2, and any set R of possible responses

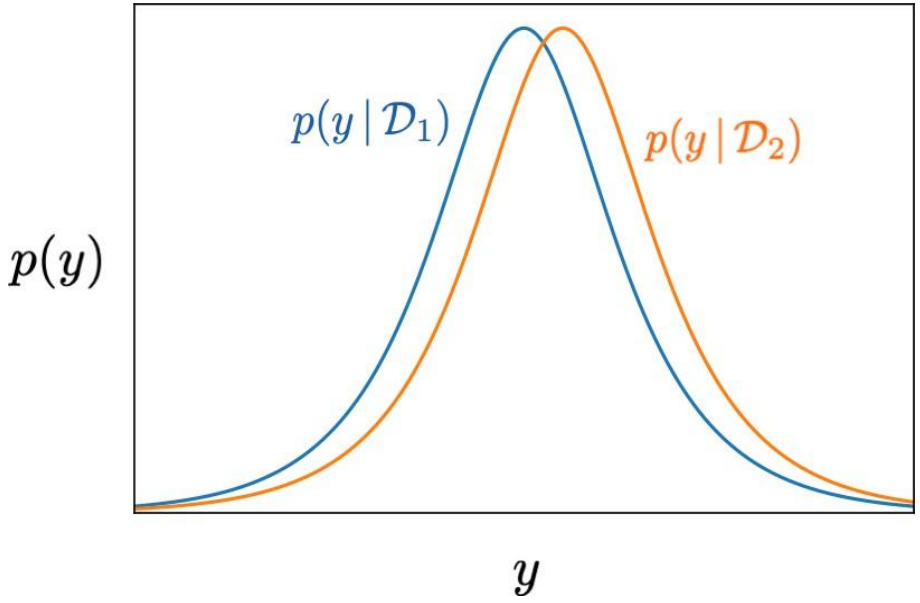
$$\Pr(\mathcal{M}(D_1) \in \mathcal{R}) \leq \exp(\varepsilon) \Pr(\mathcal{M}(D_2) \in \mathcal{R})$$

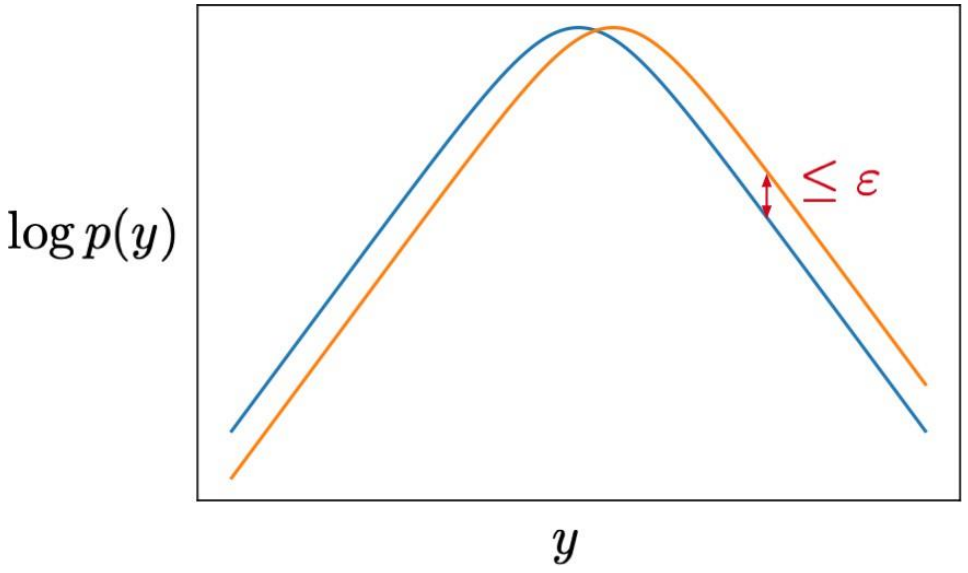
Note: for small ε, $\exp(\varepsilon) \approx 1 + \varepsilon$.

A consequence: for any possible response y ,

$$\exp(-\varepsilon) \leq \frac{\Pr(\mathcal{M}(D_1) = y)}{\Pr(\mathcal{M}(D_2) = y)} \leq \exp(\varepsilon)$$

Differential Privacy





An output from *K* that has probability 0 on dataset without your data, must also have probability 0 on dataset with your data (i.e., can't distinguish single row)

Differential Privacy

Anna is an attacker who wants to figure out if Patrick (x) is in the cancer database D. Her prior probability for him being in the database is 0.4. D is ε-differentially private. She makes a query and gets back $\mathrm{y} = \mathcal{M}(D)$

She's narrowed it down to two possible databases D_1 and D_2 , which are identical except that $x \in D_1$ and $x \notin D_2$.

After observing y , she computes her posterior probability using Bayes' Rule:

$$\begin{aligned}
\Pr(x \in \mathcal{D} \mid y) &= \frac{\Pr(x \in \mathcal{D}) \Pr(y \mid x \in \mathcal{D})}{\Pr(x \in \mathcal{D}) \Pr(y \mid x \in \mathcal{D}) + \Pr(x \notin \mathcal{D}) \Pr(y \mid x \notin \mathcal{D})} \\
&\geq \frac{\Pr(x \in \mathcal{D}) \Pr(y \mid x \in \mathcal{D})}{\Pr(x \in \mathcal{D}) \Pr(y \mid x \in \mathcal{D}) + \exp(\varepsilon) \Pr(x \notin \mathcal{D}) \Pr(y \mid x \in \mathcal{D})} \\
&= \frac{\Pr(x \in \mathcal{D})}{\Pr(x \in \mathcal{D}) + \exp(\varepsilon) \Pr(x \notin \mathcal{D})} \\
&\geq 0.4 \exp(-\varepsilon)
\end{aligned}$$

Similarly, $\Pr(x \in \mathcal{D} \mid y) \leq 0.4 \exp(\varepsilon)$ So Anna hasn't learned much about Patrick.

Differential Privacy

In what sense does this definition guarantee privacy?

Suppose a data analyst takes the result $\mathrm{y} = \mathcal{M}(D)$ and further processes it with some algorithm f (without peeking at the data itself). Is it still private?

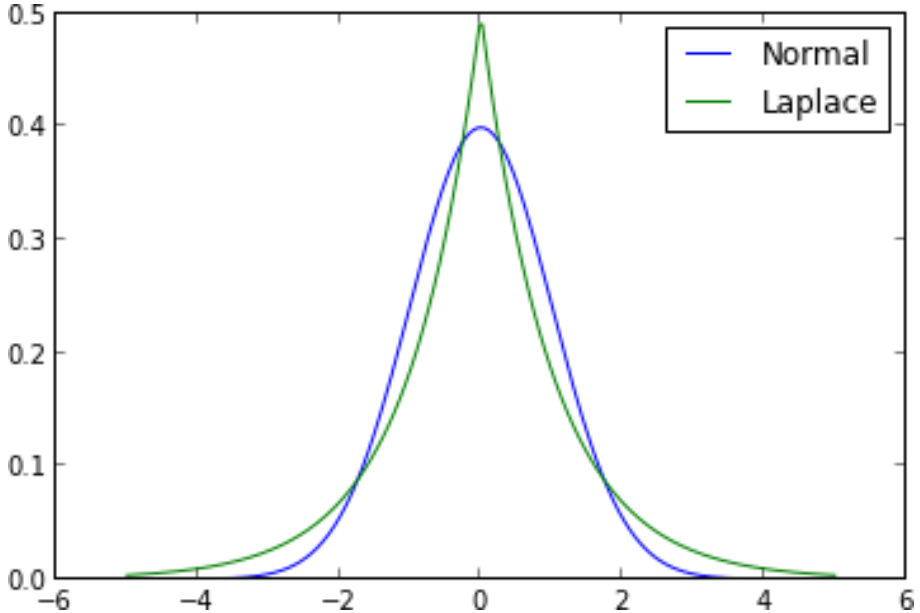
Let $\mathcal{R}$ be a set of possible outputs, and $\mathcal{R}'$ be the pre-image under f, i.e.,$\mathcal{R}' = \{y: f(y) \in \mathcal{R}\}$

$$\begin{aligned}\Pr(f(\mathcal{M}(\mathcal{D}_1)) \in \mathcal{R}) &= \Pr(\mathcal{M}(\mathcal{D}_1) \in \mathcal{R}') \\ &\leq \exp(\varepsilon) \Pr(\mathcal{M}(\mathcal{D}_2) \in \mathcal{R}') \\ &= \exp(\varepsilon) \Pr(f(\mathcal{M}(\mathcal{D}_2)) \in \mathcal{R})\end{aligned}$$

Hence, the composition $f \circ \mathcal{M}$ is also ε-differentially private. No matter how clever the analyst is, or the resources she throws at it, she can't learn more than ε about an individual entry!

Noise Injection

- When database is queried, add random noise to the answer
- Formally, if $f(x)$ is function we wanted to compute from database, we return $f(x) + C$, where C is random value
  - To satisfy differential privacy, we can determine value C using a Laplace or Normal distribution



Differential Privacy Concerns

- Querying repeatedly can undermine privacy guarantees
  - E.g., Statistical sampling techniques for mean values across groups and use group-wise correlation to detect information for specific individual.
- Randomness can cause small groups to be "washed out"
  - The noise added may too large and completely obscure any meaningful differences between the small group and the rest in the dataset.

Today's Agenda

1. Introductory Provocations
2. What is Privacy?
3. Privacy and Big Data
4. Approaches to data privacy
   - Anonymization
   - Encryption
   - Randomized Response
   - Differential Privacy
5. **Large Language Model (LLM)**